Final Model Analysis and Interpretation

ADS 503 Final Project

Vince Garcia

Libraries

```
knitr::opts_chunk$set(
    message = FALSE,
    warning = FALSE,
    include = FALSE
)
library(caret)
```

```
Warning: package 'caret' was built under R version 4.5.3

Loading required package: ggplot2

Warning: package 'ggplot2' was built under R version 4.5.3

Loading required package: lattice
```

```
library(pROC)
```

```
Type 'citation("pROC")' for a citation.


Attaching package: 'pROC'

The following objects are masked from 'package:stats':

    cov, smooth, var
```

```r
library(scales)
library(glmnet)
```

```
Warning: package 'glmnet' was built under R version 4.5.3
```

```
Loading required package: Matrix
```

```
Loaded glmnet 5.0
```

```r
library(ggplot2)
library(gbm)
```

```
Warning: package 'gbm' was built under R version 4.5.3
```

```
Loaded gbm 2.2.3
```

```
This version of gbm is no longer under development. Consider transitioning to gbm3, https://g
developers/gbm3
```

```r
library(PRROC) # Explicitly loaded for Precision-Recall evaluation metrics
```

```
Warning: package 'PRROC' was built under R version 4.5.3
```

```
Loading required package: rlang
```

```
Warning: package 'rlang' was built under R version 4.5.3
```

```r
set.seed(503)
```

Importing from Model Tuning

Model Selection & Justification

From this Proabability Calibration Curve, all models are trained on the same test data. As the data spreads out, each model improves its observed event percentage up. However, when the bin midpoint reaches 100, LDA takes a noticeable drop while GBM maintains the highest Observed Event Percentage. Highest overall performance: GBM

Highest Mean ROC: GBM Highest Mean Sensitivity: Random Forest Highest Mean Specificity: LDA

Among the 4 models tested, GBM has the highest ROC, which can be determined as the overall performance. While the ROC is a reliable measure of overall performance, the linear model, LDA does have the highest specificity while Random Forest has the highest sensitivity.

Important Variables for Winning Model

Above are the weights of each variable in the GBM model. General Health remains the most critical variable to the model with an overall % score of 100, followed by HighBP (77.06%) and BMI (44.63%).

Visual Representation of Variable Importance of GBM

The distribution of importance shows that the leading indicators of diabetes are: General Health (100%), High BP (77.06%), BMI (44.63%), Age (30.15%), HighChol (23.91%), and DiffWalk (4.96%).

Confusion Matrix

The final values for the GBM Model are: ROC AUC : 0.8302 Accuracy : 0.7504 Balanced Accuracy : 0.7504 Sensitivity : 0.7748
Specificity : 0.7260